

Weed out the laggards: Sustainability as a credit filter

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Executive Summary



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- **Financial fundamentals remain the primary drivers of credit risk.** Across more than 7,400 companies and 280,000 firm-year observations, profitability (ROA), leverage and size collectively explain the vast majority of variation in credit risk. Sustainability is statistically significant but smaller in magnitude: financial fundamentals carry coefficients 3-7 times larger than the sustainability variable in our global sample. A firm's balance sheet, earnings power and scale determine whether it can service its debt - sustainability adds an incremental but real signal on top of that financial bedrock.
- **Sustainability acts as a credit-risk filter, but its signal concentrates at the bottom of the distribution.** Weaker sustainability is consistently associated with higher default risk. A one-decile improvement in sustainability (roughly +7.5 points on a 0-100 scale) corresponds to an approximately 0.25pp reduction in default probability - a 12-25% relative reduction for firms in the worst default-risk decile. The relationship is non-linear: moving from poor to average sustainability materially improves credit outcomes; moving from average to best-in-class delivers little additional benefit.
- **Environmental performance is the clearest predictor of default risk.** A 10-point improvement in the environmental score is associated with a 0.9-point gain in the Altman Z-score - enough to move borderline firms out of the distress zone. Governance, by contrast, is the key driver of broader credit quality, consistent with financial discipline and transparency underpinning financial strength.
- **European companies lead on sustainability scores and show the greatest sensitivity to ESG factors.** Regional baselines differ sharply: Europe leads (average combined score in the 50s), the US trails (40s, with environmental scores near 30) and Japan scores well on environment (near the high-40s) but weaker on social metrics. Environmental leadership carries the strongest credit premium in emerging markets, where best-in-class issuers benefit from a scarcity premium as standards tighten. In the US, holistic ESG profiles matter more than any single pillar; in Europe, the combined score links to both credit risk and credit quality.
- **The clearest ESG-credit links cluster in sectors exposed to energy transition, operational and reputational risk - notably communication services, consumer staples and energy.** Our sector analysis finds meaningful sustainability-credit relationships in seven of eleven sectors, with the largest effects in these three. Communication/software tends to concentrate financially strong firms that can fund sustainability investment and are rewarded for risk management that encompasses ESG. Consumer staples and energy include carbon-intensive sub-sectors - textiles to oil & gas - where pollution, labor issues, carbon pricing and stranded-asset dynamics can directly hit cash flows, asset values and funding access. The relationship is weaker in materials and IT, reinforcing that sustainability integration in credit requires a sector materiality lens, not a blanket score overlay.

- **These findings are robust across data providers and confirmed by our internal credit assessments.** Replicating the analysis with alternative sustainability scores yields virtually identical results. Using our proprietary internal credit grades, we find that sustainability is already implicitly embedded in analysts' assessments - the sustainability score and environmental and social pillars are all statistically significant predictors of internal grades. Analysts are capturing sustainability-related risks in practice, a more systematic approach could sharpen that signal further.
- **The practical implication is clear: screen out the weakest.** Banks should embed sustainability as a negative screen in credit pricing and due diligence. Credit investors should use it primarily as a downside protection tool. Corporate issuers face a real credit penalty for sitting below the sustainability threshold - and no proportionate reward for exceeding it. The imperative for all market participants is the same: clear the floor. Chasing the ceiling is a marginal bonus.



Unsustainable practices can generate financial stress

Sustainability has moved from a niche concern to a mainstream factor in extra-financial evaluation of corporates over the past decade. The underlying rationale is simple: sustainability issues directly or indirectly influence the fundamentals that eventually determine a company's ability to generate revenues, make a profit or just repay debt. If a company faces environmental fines or a factory is shut down by floods or wildfires, its revenues suffer and costs rise – impeding debt service. If poor labor practices lead to strikes or a consumer boycott, cash flows can dry up. And if weak governance leads to fraud or compliance breaches, a firm can quickly find itself facing legal penalties and loss of investor confidence. All these sustainability-related events hit the company's financial position, translating into traditional credit risk and increasing the probability of default.

Climate change and environmental management are now recognized as material credit factors. Physical risks (e.g. extreme weather, droughts, rising sea levels) can destroy assets and disrupt supply chains¹. For example, in 2019, wildfires in California saddled a major utility with staggering liabilities, contributing to its bankruptcy. Transition risks are also mounting: Governments imposing carbon taxes or banning high-emission activities can strand assets and force costly business model changes². Companies heavily reliant on fossil fuels or with poor pollution controls may face higher costs, asset write-downs and weaker financial ratios, all of which hurt credit quality. On the other hand, firms proactive in cutting emissions or investing in cleaner technology can reduce regulatory risks and operating costs over time, bolstering their credit outlook.

How a company manages its workforce, customers and communities also bears on creditworthiness. Neglecting safety and labor rights, for instance, can lead to workplace accidents or strikes that interrupt production. In the mining industry, fatal accidents not only incur direct costs but can prompt license suspensions. Similarly, poor product safety or customer data breaches can trigger lawsuits and erode a firm's reputation and revenues. By contrast, companies that invest in human capital (e.g. training employees, maintaining good labor relations) and uphold product quality tend to have more resilient operations and loyal customers. Although spending on social initiatives may add costs in the short term, that stability should translate into steadier cash flows and lower credit risk in the long run. Lastly, corporate governance has long been linked to credit outcomes. From Enron's collapse to more recent corporate scandals, history shows that failures were rooted in poor governance. Lax oversight, misaligned incentives or opaque disclosure can mask mounting problems until it is too late. When governance breaks down, firms are prone to accounting fraud, corruption or reckless risk-taking that can rapidly undermine their financial position. For creditors, strong governance is a first line of defense: it means competent management, prudent financial controls and transparency. In practice, governance issues can crystalize into credit events through regulatory penalties, adverse audit findings or sudden leadership crises. Any of these can spook investors and restrict a firm's access to funding.

¹ See our previous report [on adaptation](#)

² See our previous report [on stranded assets](#)



Sustainability matters for credit quality and credit risk

Over the last ten years, a growing body of research has examined whether strong sustainability actually translates into better credit outcomes. The evidence, especially from the most recent studies, generally finds that companies with higher ESG ratings tend to have lower credit risk. On the whole, firms that excel on ESG enjoy tangible credit benefits. They often face lower borrowing costs and fewer nasty surprises. For example, multiple studies of corporate bond markets have found that companies with high ESG scores pay narrower credit spreads than peers, even after controlling for their credit ratings. In other words, bond investors demand a smaller risk premium, viewing high-ESG issuers as safer bets. These firms also exhibit lower volatility and reduced tail risk of extreme losses. The upshot is that a strong ESG profile appears to enhance financial resilience, which lenders reward with cheaper debt capital (see appendix for more references on the sustainability-credit risk nexus).

To further investigate the sustainability-credit risk nexus, we conduct an extensive statistical analysis on a global dataset of companies. Our study covers firms worldwide over the period 1991-2025, using sustainability scores from Refinitiv (see appendix for more details). This dataset spans both developed and emerging markets, with a broad sector representation. We measured credit risk in two ways: (i) the Altman Z-score, a classic metric of default risk based on financial ratios, and (ii) a credit-quality index capturing broader aspects of financial health (primarily profitability, leverage and cash flow metrics). The Altman Z-score was available for a subset of about 2,100 firms (generally larger companies with complete

financial data), while the credit quality index covered a larger sample of 7,400 firms, giving us a comprehensive view of sustainability impacts on creditworthiness. We include firms from North America, Europe, Asia-Pacific and emerging economies. Sustainability scoring coverage has expanded significantly in the past decade; still, not all companies have sustainability ratings or all the inputs for the Altman Z-score. For example, among emerging market companies in our sample, only about 44% have sustainability scores available (as coverage tends to be thinner for smaller EM firms). By contrast, in the US sample almost 67% of firms have sustainability scores, reflecting better disclosure and data coverage in the US. Developed Europe falls in between (around 65% sustainability coverage). These proportions highlight that while sustainability data is now widespread, it is not yet universal. Nevertheless, our total sample comprises over 280,000 firm-year observations, providing a rich basis to detect patterns.

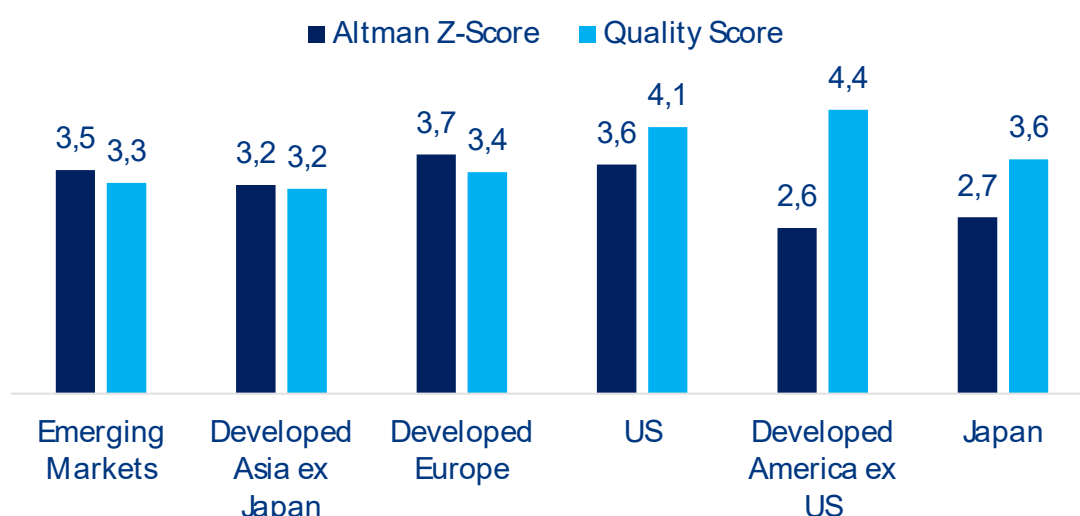
European companies boast the highest sustainability scores (the average combined score in Europe is about 50 out of 100), compared to the low 40s in emerging markets. US companies trail Europe: the average US sustainability score is around 40, with particularly low marks on environmental factors (the average US environmental pillar score is barely 30, reflecting that many American firms score poorly on carbon emissions and resource use relative to global peers). Japan, interestingly, scores well on environmental issues (average E score close to 48) but somewhat lower on social metrics. These regional differences suggest that European

businesses have led on sustainability, especially on social and environmental dimensions, while North American and some Asia-Pacific firms lag in environmental performance. Does strong sustainability in a region correspond with better credit profiles? The Altman Z-score in our sample averages around 3-4 for most regions (see Figure 1), with higher values indicating safer balance sheets³. Developed Europe had a slightly higher mean Z-score (3.7) than the US (3.6) or emerging markets (3.5). These averages put the median company safely out of distress, though the range is wide. We also construct a credit quality index for each firm-year to capture overall financial health. This composite index synthesizes profitability (return on assets) and cash-flow generation (operating cash flow to assets), interest coverage and leverage which are key inputs to credit quality. By design, higher values of this index indicate stronger credit fundamentals (high profits, robust cash flows, less debt). Not surprisingly, developed market firms tend to score better on this quality index on average, given their generally more mature, stable financials.

We find that a strong sustainability score is associated with lower credit risk. We employ panel regression analysis to quantify the relationship between sustainability performance and our credit risk measures⁴. In simple terms, when a company's sustainability score is higher, do we subsequently observe better credit outcomes? To answer this, we ran regressions of Altman Z-scores and the credit quality index on sustainability scores, controlling for other factors. Specifically, we include controls for firm size (larger firms often have more diversified risks)

and profitability (a higher return on assets naturally boosts credit metrics). We also account for fixed effects (essentially controlling for broad differences across sectors, years and regions) to isolate the impact of sustainability. Importantly, sustainability scores were lagged by one year in the models, meaning we related last year's sustainability performance to this year's credit risk indicators. This helps address causality (i.e. ensuring we are capturing sustainability influencing credit, not the other way around, and also to be consistent with the timing of availability of information, as today we know the sustainability score based on the past year's data). First of all, we need to underline that coefficients for fundamentals are significant and larger across all our regressions compared to sustainability measures. This means that financial metrics are the leading drivers of credit risk. However, we find that sustainability also matters to a significant extent: The coefficient on the Environmental Pillar Score implies that if a firm's environmental score rises by 10 points (say from 50 to 60), its Altman Z-score is expected to be about 0.9 higher on average (see Table 1). Such an effect is meaningful: given many firms' Z-scores hover in the 1.5-3.0 range, a nearly 1 point boost could move a firm from the borderline "grey zone" closer to safety. The overall ESG combined score (which integrates all pillars and accounts for controversies) had a positive coefficient of about 0.05. This suggests that, taken as a whole, strong sustainability performance is associated with better credit health (higher Altman Z-score), although among the individual components, the environmental factor is doing most of the work in explaining that link.

Figure 1: Distribution of credit risk and quality scores by regions



Sources: Refinitiv, Allianz Research

³ For context, a Z-score above 2.6 is considered "safe", while below 1.1 is considered "distressed"

⁴ See appendix for methodological details

Table 1: Distribution of credit risk and quality scores by regions

Sustainability measure vs credit risk (Altman Z-score)	Beta coefficient of the sustainability measure	Beta coefficient of fundamentals		
		Profitability	Size	Cashflow Strength
ESG Combined Score	0.05*	0.34***	0.28***	0.14***
E Pillar Score	0.09***	0.27***	0.36***	0.18***
S Pillar Score	-0.01	0.52***	0.42***	0.24***
G Pillar Score	0.02	0.49***	0.38***	0.22***

*, ** and *** denote respectively 10%, 5% and 1% significance level

Sources: Refinitiv, Allianz Research

Sustainability is also linked to broad credit quality.

When looking at credit quality, like for credit risk, fundamentals are more important. Nevertheless, we also find here that the combined ESG score has a positive and statistically significant coefficient, indicating that firms with higher overall sustainability performance tend to exhibit better credit quality. In practical terms, a 10-point increase in the ESG score corresponds to about 0.16 higher on our credit quality index – a modest but non-trivial improvement in underlying financial strength (see Table 2). Interestingly, when we break it down into pillars, we see that the governance pillar shows a positive effect. This suggests that good governance practices (board structure, minority shareholder protection, transparency, etc.) are contributing to better credit fundamentals, which is intuitively consistent with the idea that well-governed firms have more disciplined financial management. At a global level, our findings imply that sustainability matters for credit risk and credit quality, but the relative importance of E, S, and G differs between the two. For default risk (Altman Z), environmental performance is the standout (i.e. companies with stronger environmental stewardship tend to be safer from a credit risk perspective). This resonates with the notion that environmental risks (climate, pollution, regulation) have become financially material and can drive default outcomes if unmanaged. In contrast, social and governance factors did not show up in the default risk regression, possibly because their effects on default are indirect or longer-term (or already reflected in

other control variables like profitability). For credit quality, we see that having a robust overall sustainability profile helps, and there is a hint that governance (and to an extent social) is playing a role.

E matters for EMs while overall sustainability is more important for US and Europe. Looking at different regions⁵, we find that environmental pillar impacts are strongest in emerging markets when it comes to credit quality. In this group, a higher environmental score was associated with a significantly better credit quality index. Emerging market firms that are environmental leaders tend to also have stronger credit fundamentals. One interpretation is that in developing economies, where environmental standards and enforcement are rapidly tightening, firms that proactively manage environmental risks gain a credit edge – perhaps by avoiding fines or attracting sustainability-linked capital. Additionally, emerging market companies with strong sustainability ratings might enjoy a reputational premium that improves investor confidence in their financial stability. Moreover, in emerging markets, the governance pillar had a positive relationship with credit quality, suggesting governance reforms are rewarded in those countries with better perceived creditworthiness. In the US, our models showed that overall sustainability performance had a mild positive relationship with credit quality, but none of the individual E, S, G pillars were statistically significant on their own. This might indicate that US credit investors take a holistic view

⁵ North America (US and other developed Americas), Europe, Asia-Pacific (developed Asia ex-Japan, and Japan) and Emerging Markets.

– i.e. a company has to be well-rounded on sustainability to get credit recognition, and no single pillar being enough. Europe showed a positive link with the overall sustainability performance for both credit risk and credit quality. Europe's long-standing focus on sustainability could mean investors and lenders have integrated these extra-financial criteria in their analysis.

Sustainability matters most for communication services, consumer staples and energy sector credit. Overall we find significant links between sustainability and credit risk/quality for seven sectors out of 11, underlining the broad relevance of extra-financial performance for credit worthiness. However, the influence of sustainability is the largest in communication services & software, consumer staples and energy (see Figure 2). For communication

services and software, our results most likely stem from the fact that the sector includes financially strong companies that spend and invest in sustainability. Meanwhile, both consumer staples and energy sectors include carbon-intensive segments – from textiles to oil & gas – for which sustainability-related risks represent substantial financial risk, too. For instance, pollution risks and labor related-issues could disrupt firms in textiles while carbon prices and stranded assets are clear hurdles for oil & gas companies. At the other end of the spectrum, although relevant, sustainability scores seem less important for credit metrics for firms in the materials and IT sectors.

Table 2: ESG and Credit Quality Index (global panel regression)

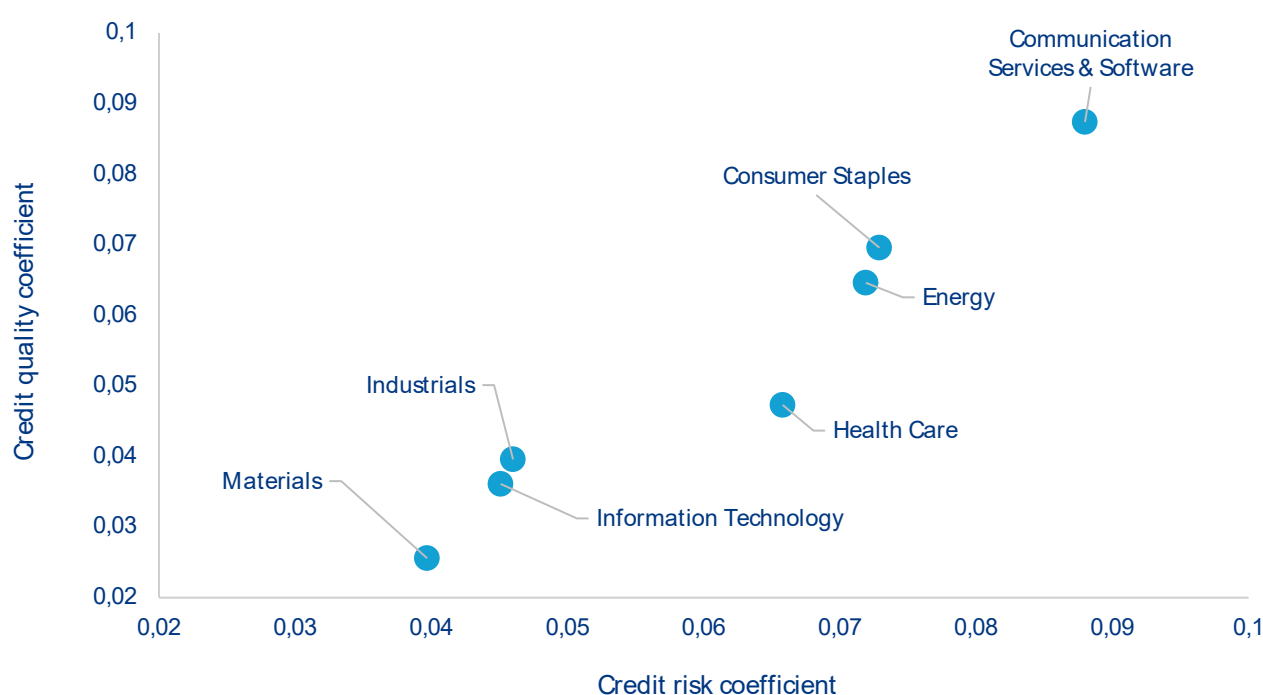
Sustainability measure vs credit quality	Beta coefficient of the sustainability measure	Beta coefficient of fundamentals ⁱ
		Size
ESG Combined Score	0.02***	0.14***
E Pillar Score	0.01	0.14***
S Pillar Score	0.01	0.26***
G Pillar Score	0.01*	0.19***

*, ** and *** denote respectively 10%, 5% and 1% significance level

ⁱ We do not include profitability and cashflow here as they are used to compute credit quality

Sources: Refinitiv, Allianz Research

Figure 2: Sustainability score coefficients for credit risk and credit quality



Sources: Refinitiv & Allianz Research – Note: we only show significant coefficients.

Our findings hold when using MSCI sustainability data instead of Refinitiv, confirming that the sustainability-credit risk nexus is not an artefact of a single data provider.

A recurring concern in the sustainability-credit literature is that results may be sensitive to the choice of rating provider, given well-documented divergences across sustainability scoring methodologies (Berg, Kölbel & Rigobon, 2022). To address this, we replicate our core Altman Z-score regression using MSCI sustainability scores in place of Refinitiv. The MSCI-based sustainability coefficient is positive and highly significant ($\beta = 0.04$, $p < 0.01$ in Table 3), closely mirroring the Refinitiv result ($\beta = 0.05$ in Table 1). All control variables – profitability, size, leverage, cash-flow strength and interest coverage – retain the expected signs and significance. This consistency across two independent data providers strengthens our confidence that the underlying relationship between sustainability performance and credit risk is genuine and not driven by idiosyncratic scoring choices.

Using our proprietary internal credit grades, we confirm that sustainability carries credit-relevant information beyond what traditional financial analysis captures.

Beyond third-party credit-risk proxies such as the Altman Z-score, we test whether the sustainability-credit link holds against our own internal credit grading system, which

runs from 1 (strongest) to 10 (weakest) and incorporates qualitative analyst judgment alongside financial metrics. Table 4 reports the results using Refinitiv sustainability scores as explanatory variables. The sustainability score, environmental pillar and social pillar all exhibit negative and highly significant coefficients ($p < 0.01$), meaning that companies with stronger sustainability performance receive better internal credit grades. The governance pillar, however, is not statistically significant – likely because governance considerations are already explicitly embedded in our internal credit assessment process. While the economic magnitudes are modest – a move from the worst to the best sustainability score corresponds to roughly a quarter-notch improvement on the 1-to-10 scale – the statistical significance validates that sustainability captures credit-relevant information incremental to what traditional financial analysis already reflects. This supports using sustainability as a complementary signal – particularly as a negative screen, as we demonstrate below – rather than as a standalone driver of credit differentiation.

Table 3: Sustainability and Altman Z-score – MSCI data (global panel regression)

Sustainability measure vs credit risk (Altman Z-score)	Beta coefficient of the sustainability measure	Beta coefficient of fundamentals			
		Profitability	Size	Leverage	Cashflow Strength
MSCI Sustainability Score	0.04***	1.29***	0.10***	-1.06***	0.26***

*, ** and *** denote respectively 10%, 5% and 1% significance level

Sources: Refinitiv, Allianz Research

Table 4: Refinitiv sustainability scores and internal credit grades

Sustainability measure vs internal credit grade	Beta coefficient	Significance
Sustainability Score	-0.04	***
Combined Sustainability Score	-0.03	***
Environmental Pillar Score	-0.03	***
Social Pillar Score	-0.04	***
Governance Pillar Score	-0.01	-

Negative coefficients indicate better (lower) credit grades. All regressions control for firm size.

*, ** and *** denote respectively 10%, 5% and 1% significance level. Sources: Refinitiv, Allianz Research

When we split firms into low, medium and high sustainability groups, the credit-risk relationship is concentrated in the bottom tercile – suggesting sustainability is most actionable as a filter to exclude the worst-scoring companies. To test whether the sustainability-credit link is linear or concentrated in specific segments, we estimate a spline regression that allows the slope to differ across sustainability terciles (using MSCI data). The results are striking (Table 5): the coefficient for the low sustainability group is 0.10 and highly significant at the 1% level, implying that among firms with the weakest sustainability profiles, each additional point of improvement is associated with a meaningful increase in the Altman Z-score. By contrast, the coefficients for the medium (0.01) and high (0.02) groups are small and statistically insignificant. In other words, the marginal credit benefit of better sustainability is strongest at the bottom of the distribution and tapers off once firms reach a moderate standard. This non-linearity has a clear practical implication: rather than rewarding top sustainability performers, the data supports using sustainability primarily as a negative screen – flagging or excluding firms in the lowest sustainability tercile, where poor practices are most strongly associated with elevated credit risk.

Translating our results into probability-of-default terms, a one-decile improvement in sustainability corresponds to a reduction in default probability of approximately 0.25pp. To make our findings directly actionable for

credit practitioners, we link the sustainability-to-Altman Z relationship to probability of default (PD) through a two-step estimation. First, from our MSCI regression, a one-decile improvement in sustainability score (approximately 7.5 points) is associated with a 0.29-point increase in the Altman Z-score. Second, using a logit regression of observed PD on winsorised Altman Z-scores ($\beta = -0.0096$, $p < 0.01$), we compute the implied change in default probability. The resulting chain implies that a one-decile sustainability improvement – approximately 7.5 points on the 0–100 sustainability scale, for example moving from a score of 12 to 20 for firms in the bottom decile – translates to approximately a 0.25pp reduction in PD. To put this in perspective, observed default probabilities in our sample range from near zero to around 5%. For a company in the worst decile of default risk, where PDs typically reach 1–2% or higher, a 0.25pp reduction represents a 12–25% relative improvement in creditworthiness – a meaningful shift that can alter risk classification. The effect compounds across large credit portfolios and is substantially larger for the environmental pillar using Refinitiv data: the stronger E-pillar coefficient ($\beta = 0.09$ per point, from Table 1) implies that a 10-point improvement in environmental performance corresponds to an even greater PD reduction – underscoring that environmental risk management is the single most credit-relevant dimension of sustainability.

Table 5: Sustainability tercile effects on Altman Z-score – Spline regression (MSCI data)

Sustainability group	Beta coefficient	Significance
Low (bottom tercile)	0.10	***
Medium (middle tercile)	0.01	-
High (top tercile)	0.02	-

Controls: Profitability, Size, Leverage, Cash-flow Strength, Interest Coverage – all significant at 1%.

Sources: MSCI, Allianz Research



Implications for financial institutions, investors and businesses

Commercial banks should embed the floor; pricing the premium is about conviction. The core finding for commercial banks is specific: Borrowers in the bottom sustainability tercile carry statistically higher default risk, and the 0.25pp PD reduction per sustainability decile is material in credit loss terms at portfolio scale. The practical response is to embed sustainability as a negative screen in credit due diligence: flagging and re-pricing exposure to low-sustainability borrowers, rather than rewarding high sustainability credentials with lower margins across the board. A borrower moving from moderate to strong sustainability does not generate a proportionate reduction in expected credit loss but a borrower in the bottom tercile generates elevated risk that traditional financial analysis may not fully surface. In practice, this integration means incorporating sustainability risk flags into internal ratings and pricing grids, particularly for borrowers in carbon-intensive sectors and those with governance deficiencies.

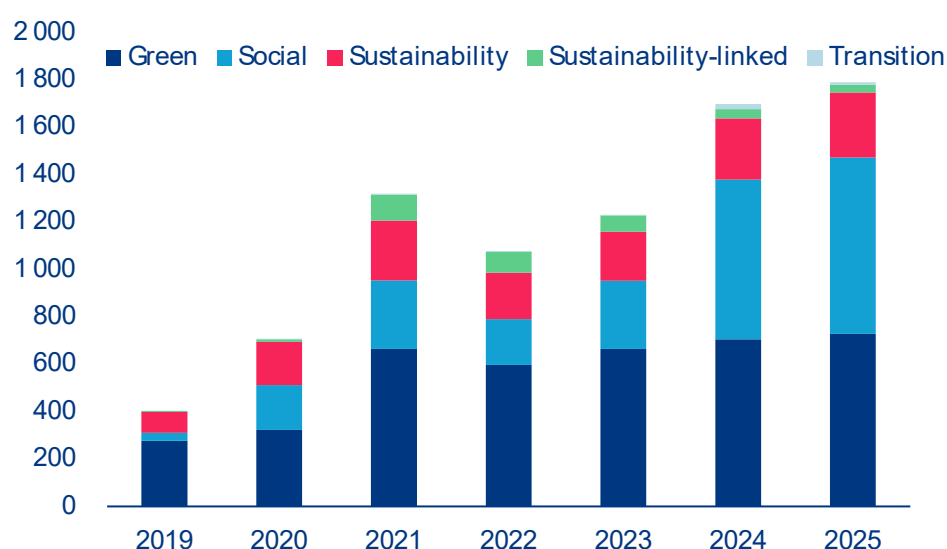
Broad-based credit investors should consider ESG filters as downside protection, not necessarily an alpha generator. For asset managers and credit investors, our findings support sustainability integration primarily as a downside risk mitigation tool. Indeed, the credit signal concentrates in identifying default-prone issuers, not in predicting outperformance by top-rated ones. A portfolio that systematically screens out bottom-tercile sustainability issuers captures the bulk of the credit risk

reduction; however, tilting toward the highest-rated sustainability issuers does not generate proportionate improvement in expected risk-adjusted returns. This reframes the analytical objective: the primary use case is loss avoidance, not return enhancement. This framing is consistent with fund-level evidence that sustainability-tilted credit portfolios have exhibited lower spread volatility and drawdowns during stress periods – precisely because the downside-protection effect is real and concentrated at the weakest issuers. Forward-looking sustainability indicators – transition strategies, green capex trajectories, emissions commitments – are increasingly relevant as transition risk begins to crystallize in credit spreads, particularly in the three sectors where our analysis shows the largest effects. Issuer engagement focused on sustainability risk management in communication services, consumer staples and energy represents a more targeted and defensible application of the credit signal than broad portfolio sustainability tilts. Investors who can distinguish genuine sustainability improvement at the bottom of the distribution from cosmetic score improvements elsewhere will have the most informative signal as the market matures.

Insurers are exposed to sustainability-related credit risk through both underwriting and investment activities and the sector concentration in our findings is directly relevant to where analytical effort should be focused. The credit signal from sustainability is largest in communication services, consumer staples and energy, and weakest in materials and IT. A uniform sustainability overlay applied to a corporate bond portfolio overstates the signal in low-materiality sectors and risks underweighting it in high-materiality ones – concentrating integration on the three high-signal sectors generates a materially better signal-to-noise ratio. The analytical focus for insurers is also shifting from static ESG scores toward forward-looking indicators to distinguish issuers that are credibly managing transition risk from those accumulating deferred exposure. During phases of spread decompression, when idiosyncratic credit risk dominates and return dispersion rises, sustainability-integrated analysis is most valuable precisely because the signal concentrates at the bottom of the distribution: it helps identify which issuers are deteriorating while others hold, rather than providing a uniform quality tilt. As climate change and shifting social expectations reshape sectoral risk profiles over the medium term, sector-level sustainability views should feed directly into strategic allocation decisions – the large dispersion our analysis shows across sectors in how strongly sustainability affects credit risk is itself a portfolio signal.

For corporations, improving sustainability can yield tangible benefits in terms of credit quality and funding costs. Our findings translate into a specific and financially grounded strategic priority for corporate issuers: the credit and financing benefits of sustainability concentrate at the threshold, and the penalty for falling below it is real. Companies with weak sustainability profiles (particularly weak environmental practices and poor governance) face measurably higher borrowing costs, tighter capital market access and elevated default risk. The strategic implication is to prioritize the most material sustainability risks rather than optimizing for the highest possible combined ESG score. Clear, measurable commitments to addressing material environmental and governance risks, transparent reporting and credible transition strategies are the inputs that matter for credit assessment. Companies that credibly clear the sustainability floor benefit from improved credit terms and a broader investor base; those that do not will face systematic pricing penalties as banks and investors formalize negative screen frameworks. The ESG investment that removes a credit penalty has a higher financial return than the ESG investment that earns marginal recognition at the top of a distribution where the signal is weak.

Figure 3: Labelled bond issuance volume (in USD bn)



Sources: Bloomberg. Includes corporate, government, supranational bonds, municipals and securitized products

Figure 4: 12M Normalized OAS

Source: Bloomberg

The expanding market for sustainability-labelled bonds provides investors with targeted exposure to environmental and social objectives, supported by use-of-proceeds and reporting frameworks that enhance transparency and allow for an assessment of issuers' sustainability trajectories. Issuance of GSSS bonds has risen markedly over the last six years, driven largely by issuers with comparatively strong sustainability profiles (see Figure 3). It remains dominated by green bonds, with activity concentrated in the financials and utilities sectors. While corporate green bonds historically traded at a premium ("greenium") due to scarcity amid strong demand, this premium has largely vanished at the aggregate index level as the market has scaled up, though sector-level dispersion persists (see Figure 4). Empirically, green bonds have exhibited lower spread volatility during market widening phases than the broader corporate bond universe, suggesting potentially more resilient risk-return characteristics, provided issuers' sustainability commitments are credible.

Appendix: Data and methodology details

Refinitiv Sustainability Scores – Methodology:

Refinitiv sustainability scores are constructed using a transparent, rules-based methodology that aggregates approximately 180–200 publicly available environmental, social, and governance indicators into standardized measures of corporate sustainability performance. Raw indicators, derived from company disclosures and publicly reported information, are mapped into defined sustainability categories and themes, normalized on an industry-relative basis, and combined using pre-determined, industry-specific weights to produce Environmental, Social, and Governance pillar scores on a 0–100 scale. In addition to the standalone ESG score, Refinitiv calculates an ESG Combined (ESGC) score, which incorporates an overlay for sustainability-related controversies by adjusting the ESG score based on the severity and frequency of documented incidents. Both ESG and ESGC scores are updated as new disclosures and controversy information become available.

Data sources and sample construction: We compiled firm-level data from Refinitiv covering the years 2011 through 2024. The ESG data comes from Refinitiv's ESG database, which provides an ESG Combined Score (a comprehensive rating including environmental, social, governance aspects and deducting for controversies) as well as separate pillar scores for Environmental, Social, and Governance (each scaled 0-100, with higher = better ESG performance). We also collected financial data from Refinitiv for the same firms, including balance sheet and income statement items required to calculate the Altman Z-score and other financial ratios. Our initial universe included over 130,000 companies globally, but many were missing ESG or financial data. We restricted the sample to firm-year observations with non-null values for the variables of interest. The final panel comprises about 280,000 observations on ~7,400 unique firms worldwide. The regional breakdown of firms (by headquarters) was: roughly 18% North America, 20% Developed Europe, 6% Japan, 7% other developed Asia-Pacific, 34% Emerging Markets, and 15% unclassified/unknown region (the latter are cases where region wasn't specified, often smaller frontier market firms). Data coverage varied, with developed markets generally having better ESG disclosure. For instance, 67% of U.S. firm-year observations had an ESG score available, versus ~44% in Emerging Markets, as noted earlier.

Altman Z-score calculation: We used the **Altman Z-score** as a proxy for credit risk or default risk of a firm. The Altman Z-score is a well-known composite financial indicator developed by Edward Altman, which combines five accounting ratios using fixed weights. We applied the version for publicly traded firms (Altman's Z-score for manufacturing, since most of our firms fall in non-financial sectors). The formula is:

$$Z = 1.2 \times \frac{\text{Working Capital}}{\text{Total Assets}} + 1.4 \times \frac{\text{Retained Earnings}}{\text{Total Assets}} + 3.3 \times \frac{\text{EBIT}}{\text{Total Assets}} + 0.6 \times \frac{\text{Market Value of Equity}}{\text{Total Liabilities}} + 1.0 \times \frac{\text{Sales}}{\text{Total Assets}}.$$

This formula produces a single score where higher values indicate a lower probability of bankruptcy. As mentioned, $Z > 2.6$ is usually considered "safe", $Z < 1.1$ is "distressed", and the 1.1–2.6 range is a grey zone. We winsorized extremely high or low Z-scores to avoid undue influence of outliers (e.g., a few observations had negative equity or abnormally large market capitalizations leading to very high Z).

In our dataset, not all firms had the market value data needed (especially if not publicly traded or if data was missing), which is why the Altman Z sample is smaller (~2,100 firms). Notably, financial sector firms (banks, insurance) are excluded from Altman Z analysis since the Z-score isn't designed for them and their balance sheets aren't comparable.

Credit quality composite: To measure credit quality more broadly (and to include more firms), we constructed a composite index reflecting a firm's earnings and cash flow capacity, which are crucial to creditworthiness. Specifically, we took three indicators: Return on Assets (ROA = Net Income / Total Assets), Leverage (Total Debt to Total Assets) and Operating Cash Flow to Total Assets (OCF/TA), which capture profitability and cash generation respectively. We standardized these within each year (to account for economic cycles) and took an average to form a composite score. This "credit quality index" is not a standard metric like Altman Z, but it serves as a general gauge of a firm's fundamental strength to service debt – high values mean the company is profitable and generating cash relative to its asset base (implying it likely can cover interest and reinvest, hence higher credit quality). We included this second measure because credit risk is multifaceted; a firm might have a low Z-score due to high market leverage but still strong cash flows to pay debts, or vice versa. The composite helps capture dimensions the Z-score might miss and greatly expands coverage to smaller firms that don't have market value data.

Regression specifications: We estimated panel regressions of the form:

$$\text{CreditMetric}_{i,t} = \alpha + \beta \times \text{ESGMetric}_{i,t-1} + \gamma \mathbf{X}_{i,t-1} + \text{FE} + \varepsilon_{i,t},$$

where $\text{CreditMetric}_{i,t}$ is either Altman Z or the credit quality index for firm i in year t . $\text{ESGMetric}_{i,t-1}$ is the ESG variable of interest (we ran separate regressions for the Combined ESG Score, ESG Score, Environmental pillar, Social pillar, and Governance pillar) lagged one year. $\mathbf{X}_{i,t-1}$ represents control variables, which included firm size (log of total assets) and profitability (ROA) – both lagged one year as well. “FE” denotes fixed effects; for the Altman Z models, we included sector-by-year fixed effects (SYFE), meaning we account for any shocks or trends affecting a whole sector in a given year (like an oil price spike affecting all oil & gas firms’ Z-scores in 2020, for example). In the credit quality models, we included both sector-year and region-year fixed effects (since that sample was much larger, we could control for regional macro conditions as well). These fixed effects absorb broad macroeconomic or industry effects, isolating the impact of ESG at the firm level. We did not include firm fixed effects because our ESG scores don’t exhibit enough within-firm time variation for many firms (many have short time series), and we aimed to capture between-firm differences as well – effectively assessing if higher-ESG firms have better credit metrics than lower-ESG firms, controlling for size and ROA.

We used robust standard errors clustered at the firm level to account for heteroskedasticity and serial correlation. The statistical significance was assessed at the 10%, 5%, and 1% levels as customary. The regression tables in the main text (Table 1 and 2) summarize the coefficient estimates for the ESG variables and their significance. The R-squared (within) for the Altman Z regressions were around 0.05 (i.e., 5% of the variation in Z within sectors/years was explained by ESG and controls – not surprising as many firm-specific factors drive Z). For the credit quality regressions, the within R-squared was higher (~0.69) because profitability (ROA) is itself part of the dependent variable and explains a lot of its variation, and fixed effects soak up a lot of variance as well.

Supplementary tables: Below we provide a summary comparison of regional results mentioned in the text for the credit quality regressions (coefficients of ESG variables by region), to give a fuller picture:

Table A1. ESG coefficient estimates by region – Credit Quality Index regressions (signs and significance of ESG Combined Score and Pillar effects)

Beta Coefficients of ESG vs Credit Risk	ESG Combined Score	E Pillar Score	S Pillar Score	G Pillar Score
US	positive & significant	positive & significant	positive	positive & significant
Developed Europe	positive & significant	positive & significant	positive & significant	positive & significant
Japan	positive & significant	positive & significant	positive	positive & significant
Developed America ex US	positive & significant	positive & significant	positive	positive
Developed Asia ex Japan	positive & significant	negative	negative	negative
Emerging Markets	negative	positive	positive	positive

Sources: Refinitiv & Allianz Research

References:

Date	Publication Source	Authors	Title	Main Findings
2011	Journal of Banking & Finance	Goss & Roberts	The impact of corporate social responsibility on the cost of bank loans	CSR concerns linked to higher loan spreads and tighter terms; lenders price ESG-related risk, especially for weaker borrowers.
2013	Journal of Business Ethics	Attig et al.	Corporate Social Responsibility and Credit Ratings	Higher CSR performance associated with better credit ratings, robust to endogeneity concerns.
2014	The Financial Review	Oikonomou et al.	The Effects of Corporate Social Performance on the Cost of Corporate Debt and Credit Ratings	Strong social performance linked to lower bond yield spreads and higher assessed credit quality.
2017	Principles for Responsible Investment (PRI)	PRI	Why ESG factors matter in credit risk analysis	ESG affects default risk, downgrade risk, and recovery values rather than just bond pricing.
2020	Banque de France / ACPR	Billio et al.	Inside the ESG ratings: (Dis)agreement and performance	Large ESG rating disagreement implies ESG–credit results depend strongly on provider choice.
2022	Review of Finance	Berg, Kölbel & Rigobon	Aggregate Confusion: The Divergence of ESG Ratings	Low correlation across ESG ratings; measurement noise complicates credit-risk inference.
2022	Journal of Risk Finance	Barth, Hübel & Scholz	ESG and corporate credit spreads	Higher ESG scores associated with lower CDS spreads; ESG shocks are priced in credit markets.
2022	NY Fed Staff Report	Seltzer, Starks & Zhu	Climate Regulatory Risk and Corporate Bonds	Climate regulatory risk causally worsens credit ratings and increases yield spreads.
2022	World Bank	Gratcheva et al.	Credit Worthy: ESG Factors and Sovereign Credit Ratings	Clarifies how ESG factors relate to sovereign creditworthiness; warns against mechanical ESG–rating mapping.
2023	International Review of Economics and Finance	Lian et al.	How does ESG performance affect bond credit spreads?	Better ESG linked to lower bond spreads via reduced financial risk and better transparency.
2025	International Review of Economics and Finance	Chen et al.	Impact of ESG factors on credit ratings	Environmental factors positively associated with European banks' credit ratings.

A photograph showing a variety of hands of different skin tones stacked on top of each other, resting on a rough, textured tree trunk. The background is a lush green forest with sunlight filtering through the leaves. The text 'Our team' is overlaid on the image, with 'Our' in white and 'team' in yellow.

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